

TASK1: RESEARCH ON WORKFLOW MANAGEMENT SYSTEM

Workflow and Workflow management system definition.

A workflow is the way that a particular type of work is organized, or the order of the stages in a particular work process. A Workflow Management System is a system that defines, manages, and executes workflows using software based on the workflow logic. It automates the executions of applications and is robust against performance and failures.

Difference between a workflow definition and a running workflow instance.

A workflow definition is the static blueprint or template that outlines the rules, steps, and logic of a process, while a running workflow instance is the actual active execution of that blueprint for a specific task or event.

Representation of activities or work items.

In workflow systems, activities are usually modeled as named process steps with control-flow links, data inputs and outputs, resource bindings, and sometimes application calls, while work items are the runtime units offered to a person, device, or service for one specific case.

Way in which work items are assigned and routed.

Work items are assigned and routed based on rules defined in the workflow definition, which are then executed for specific users or systems within the running workflow instance.

How approvals, rejection and escalation are handled.

Approvals, rejections, and escalations are handled in a running workflow instance by executing conditional gating and timing rules predefined in the workflow definition.

How participants, roles and responsibilities are represented.

Participants are modelled as resources, meaning the humans or systems that do work, and roles group those resources so that work can be offered to a role instead of a named person. Responsibility attaches when a work item is allocated to a specific resource, who is then accountable for it and can pass it to a delegate (Russell *et al.*, 2004).

How dependencies, events and failures are handled.

Workflow systems typically treat dependencies as explicit execution constraints, events as signals that change process state or trigger actions, and failures as exceptions that require rollback, compensation, alternative routing, or recovery workflows.

How workflow systems interact with external systems.

Workflow systems interact with external systems through standard interfaces, and the engine itself stays separate from them. The WfMC Reference Model says the engine interacts with the external resources needed to carry out activities (WfMC, 1993), and it defines interfaces for process definition tools, for interoperability between engines, for end-user client tools and for administration and monitoring tools (WfMC, 1996).

How workflow history or audit information is recorded.

Workflow history is recorded by the workflow engine, which writes an audit record every time something happens. Each record names the event and has a process instance ID, that links all the records of one case, and an activity instance ID that is unique to that case. Those records are kept together with process and role data, so the history of who did what can be checked against the process definition.

How running workflows are monitored and reported on.

Running workflows are monitored by capturing process state, task status, timestamps, and execution events as work progresses, then exposing that information through a central layer for users and tools. Reporting usually sits on top of that monitoring layer as real-time dashboards, alerts, and exportable summaries tailored to operational and managerial decisions.

Reasoning on why workflow engines are reusable across different processes.

It is reusable because the process logic is kept out of the engine and stored in a process definition that the engine reads and runs. The engine only provides the generic parts and everything else that differs between processes is configured in the definition. Therefore, the same engine runs a different process just by loading a different definition.

What information belongs to a workflow definition and to a running instance?

The definition holds the reusable template: the activities, procedure rules, control data, sub-process references, per-activity settings and transition rules. The running instance holds what belongs to one case: its own data and variables, case's current state and of each activity, the task instances with who took them, and the audit records of what happened.

How real nested activities or sub-processes are modeled in real workflow systems.

They are modeled as a separately defined process that a parent definition can refer to, so it makes up part of the overall process. A sub-process therefore behaves like a single activity from the parent's point of view, while being a full process underneath. Links to parent via the root process instance ID.

Decisions that are normally configurable rather than hard-coded into the workflow engine.

The decisions that vary between organisations are kept in the process definition or in configuration, not in the engine's code. These are activities and the order they run in, the transition conditions and flow rules, and the start and termination conditions of a process.

How are changes to workflow definitions handled while work is already in progress.

Changes to workflow definitions while work is already in progress are typically handled through parallel versioning, where existing running instances are safely isolated and left to finish using their original blueprint version. Simultaneously, the engine routes all incoming new traffic to the modified definition.

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